

(7) and (9) gives

$$\begin{aligned}
u_n^{(1)} &= u_{n-3}^{(1)} + u_{n-7}^{(2)} + u_{n-11}^{(3)} + u_{n-8}^{(2)} + u_{n-12}^{(3)} \\
&= u_{n-3}^{(1)} + u_{n-7}^{(2)} + u_{n-8}^{(2)} + u_{n-5}^{(1)} - u_{n-8}^{(1)} \\
&= u_{n-3}^{(1)} + u_{n-5}^{(1)} - u_{n-8}^{(1)} + u_{n-7}^{(2)} + u_{n-8}^{(2)}
\end{aligned} \tag{10}$$

and (6) and (7) gives

$$\begin{aligned}
u_n^{(2)} &= u_{n-1}^{(1)} + u_{n-6}^{(2)} + u_{n-10}^{(3)} \\
&= u_{n-1}^{(1)} + u_{n-6}^{(2)} + u_{n-5}^{(2)} - u_{n-6}^{(1)} \\
&= u_{n-1}^{(1)} - u_{n-6}^{(1)} + u_{n-5}^{(2)} + u_{n-6}^{(2)}.
\end{aligned} \tag{11}$$

Finally, (10) and (11) gives

$$\begin{aligned}
u_n^{(1)} &= u_{n-3}^{(1)} + u_{n-5}^{(1)} - u_{n-8}^{(1)} + u_{n-8}^{(1)} - u_{n-13}^{(1)} + u_{n-12}^{(2)} \\
&\quad + u_{n-13}^{(2)} + u_{n-9}^{(2)} - u_{n-14}^{(1)} + u_{n-13}^{(1)} + u_{n-14}^{(2)} \\
&= u_{n-3}^{(1)} + u_{n-5}^{(1)} + u_{n-9}^{(1)} - u_{n-13}^{(1)} - u_{n-1}^{(1)} + u_{n-5}^{(1)} - u_{n-8}^{(1)} \\
&\quad - u_{n-10}^{(1)} + u_{n-13}^{(1)} + u_{n-6}^{(1)} - u_{n-9}^{(1)} - u_{n-11}^{(1)} + u_{n-14}^{(1)} \\
&= u_{n-3}^{(1)} + 2u_{n-5}^{(1)} + u_{n-6}^{(1)} - u_{n-8}^{(1)} - u_{n-10}^{(1)} - u_{n-11}^{(1)}
\end{aligned}$$

which gives us a universal characteristic polynomial $x^{11} - x^8 - 2x^6 - x^5 + x^3 + x + 1$. However, it factorizes as

$$x^{11} - x^8 - 2x^6 - x^5 + x^3 + x + 1 = (x - 1)(x^4 + x^3 + x^2 + x + 1)(x^6 - x^3 - x - 1),$$

and we can check whether some proper divisor is also a universal characteristic polynomial. Specifically, we can let $j \in \{1, \dots, 13\}$ and set $u_3^{(j)} = 1$ and $u_3^{(i)} = 0$ for $i \neq j$. We set the initial values for $n = k$, as (1) is only defined for $n \geq k$. It turns out that $u_{13}^{(i)} - u_{10}^{(i)} - u_8^{(i)} - u_7^{(i)} = 0$ for all $i \in \{1, \dots, 13\}$ regardless of j . By Lemma 3.2, we then have $u_n^{(i)} - u_{n-3}^{(i)} - u_{n-5}^{(i)} - u_{n-6}^{(i)} = 0$ for all $n \geq 13$ for all i , and since any set of initial values is a linear combination of the ones we have tried out, $x^6 - x^3 - x - 1$ is a universal characteristic polynomial on (ξ, I_3) . Since it is irreducible and ξ is nontrivial, it must be minimal.

As an example of an execution of the flowchart in Algorithm 1, let $v = (4, 16, \{(L_2, L_3), (R_1, R_2)\})$. The subgraph $\rho(v) \subset \sigma_3$ consists of the SCCs